Cycle Test 2: worked solutions

15 Oct 2025 · 25 marks · 4:00–5:30 PM · CEPC14

Part A

Q1

3 marks

> An M35 grade concrete is laid on a roof slab of the new civil annex building. Cube samples were made during the laying of the slab and tested after 7 and 28 days of curing. Based on the results, comment on the strength and acceptability as per IS 456.

Specimen	7-day load (kN)	28-day load (kN)
CA 15/9/24 SP_1	550	730
CA 15/9/24 SP_2	665	950
CA 15/9/24 SP_3	635	850

Standard cubes are **150 mm**, so area = 150 × 150 = 22,500 mm^2. Strength = load ÷ area.

Specimen	7-day (MPa)	28-day (MPa)
SP_1	550,000 / 22,500 = 24.44	730,000 / 22,500 = 32.44
SP_2	29.56	42.22
SP_3	28.22	37.78
Test result (mean of 3)	**27.41**	**37.48**

1. Validity of the sample (IS 456 Cl. 15.4): individual specimens must lie within **±15%** of the mean.

- 28 days: 32.44 is −13.4%, 42.22 is +12.6%, 37.78 is +0.8% → **valid** (but the spread of about 10 MPa shows poor uniformity).

- 7 days: −10.8%, +7.8%, +3.0% → valid.

2. Early strength: 7-day strength is normally about 65–70% of the 28-day value. 27.41 / 37.48 = **73%**, and 27.41 > 0.67 × 35 = 23.5 MPa → strength development is normal.

3. Acceptance (IS 456 Table 11, M20 and above):

- **Individual test result ≥ f_{ck} − 4 = 31 MPa**: 37.48 ≥ 31 → **satisfied**.
- **Mean of 4 non-overlapping consecutive results ≥ f_{ck} + 0.825 s or f_{ck} + 4 (whichever is greater)** = 35 + 4 = 39 MPa (or 35 + 0.825 × 5 = 39.1). Only one result is available; 37.48 is **below 39**, so if later results stay at this level the slab concrete would **not meet the mean criterion**.

Comment: the result exceeds the characteristic strength (37.5 > 35 MPa) and passes the individual criterion, so the concrete is **acceptable for now**. But it is only just above f_{ck} and the cubes vary widely, so the producer should **improve quality control** (batching, compaction, curing), and more samples are needed (IS 456 sampling frequency) to judge compliance with the mean criterion.

Q2

7 marks

> Design an M50 concrete mix for the foundation of bridge members. Concentration of sulphate in soil is 3. Choose the appropriate type of cement and design. Specific gravity of fine aggregate, coarse aggregate and cement are 2.65, 2.8 and 3.1. Water absorption of coarse and fine aggregate is 1.5%. Coarse aggregate of 20 mm rounded aggregate is only available. Concrete is placed by pumping (take slump according to pumping). Total moisture content in CA is 3% and in FA is 1.5%. Specific gravity of superplasticizer is 1.35.

Choice of cement: "sulphate concentration 3" is taken as **Class 3** of IS 456 Table 4 (SO_3 in soil 0.5–1.0%). For Class 3 the options are supersulphated cement or **sulphate resisting cement (SRC, IS 12330)** with min. cement 330 kg/m^3 and max. w/c 0.50, or PPC/PSC with min. 350 kg/m^3 and max. w/c 0.45. **SRC is chosen** (low C_3A resists sulfate attack).

A foundation in sulfate-bearing soil is also **"very severe" exposure** (IS 456 Table 3: buried in aggressive sub-soil): **max w/c 0.45, min cement 340 kg/m^3, min grade M35**.

(If the soil actually has **3% SO_3**, that is Class 5: SRC with a protective coating, min. cement 400 kg/m^3, max. w/c 0.40.)

Assumptions: good supervision; zone II sand; superplasticizer 1% of cement with 20% water reduction; entrapped air 1%; slump **100 mm** for pumping.

Step	Working	Result
1. Target strength	max(50 + 1.65 × 5, 50 + 6.5)	**58.25 MPa**
2. w/c	Fig. 1 of IS 10262:2019 → about 0.32 (< 0.45 ✓)	**0.32**
3. Water	186 − 25 (rounded aggregate) = 161; +6% for 100 mm slump = 170.7; −20% (SP)	**136.5 kg/m^3**
4. Cement	136.5 / 0.32 = 426.6 (≥ 340, ≤ 450 ✓)	**427 kg/m^3**
5. SP	1% × 426.6	**4.27 kg/m^3**
6. CA volume fraction	0.62 + (0.50 − 0.32)/0.05 × 0.01 = 0.656; × 0.9 for pumping	**0.590** (FA 0.410)

Mix calculations (per m^3):

Item	Volume (m^3)
Entrapped air	0.010
Cement = 426.6 / (3.1 × 1000)	0.1376
Water = 136.5 / 1000	0.1365
SP = 4.27 / (1.35 × 1000)	0.0032
All-in aggregate = 1 − (sum)	**0.7127**
CA (SSD) = 0.7127 × 0.590 × 2.8 × 1000	**1178 kg**
FA (SSD) = 0.7127 × 0.410 × 2.65 × 1000	**774 kg**

Moisture correction:

- CA: total moisture 3% − absorption 1.5% = **1.5% free water** → adds 0.015 × 1178 = 17.7 kg water; weigh **1178 × 1.015 = 1196 kg** of wet CA.
- FA: total moisture 1.5% = absorption 1.5% → **SSD, no correction**: 774 kg.
- Water to add = 136.5 − 17.7 = **118.9 kg**.

Final mix per m³ (site): SRC **427 kg**, water **119 kg**, FA **774 kg**, CA 20 mm **1196 kg**, SP **4.3 kg**. Free w/c = 0.32; proportion (SSD) **1 : 1.81 : 2.76**. Confirm by trial mixes for slump retention through the pump.

Q3

2 marks

> Define creep and shrinkage of concrete.

- **Creep**: the **gradual increase in strain with time under a sustained (constant) load**. Water in the C-S-H is forced to move by the stress. Under constant strain it shows up as **relaxation** of stress.
- **Shrinkage**: the **reduction in volume of concrete with time, without any load**, caused by loss of water (drying, self-desiccation), cooling or carbonation. Types: plastic, autogenous, drying, thermal, carbonation.

Both are time-dependent, partly irreversible, and increase with paste content and w/c.

Q4

2 marks

> Define workability of concrete. What are the tests to find workability? Why?

Workability is the ease and homogeneity with which fresh concrete can be **mixed, placed, compacted and finished** without segregation.

Tests: **slump test**, **compacting factor test**, **Vebe test**, **flow table test**, **Kelly ball penetration test**.

Why several tests: no single test measures workability directly. Each suits a range: slump for medium–high workability (site test); compacting factor for low workability (lab, more sensitive); Vebe for very dry mixes; flow table for

flowing/SP concrete; Kelly ball for quick in-place checks. Workability is checked to ensure **full compaction (5% voids → ~30% strength loss)** and batch-to-batch consistency.

Q5

2 marks

> Explain the five phases of the construction project life cycle with suitable examples.

1. **Initiation (conception)**: identify the need, feasibility study, approval. *Example: a university decides it needs a new civil annex; feasibility and budget are approved.*
2. **Planning**: scope, design, drawings, estimates, schedule, procurement plan, risk and quality plans. *Example: architect's drawings, BOQ, tendering, bar chart/CPM schedule.*
3. **Execution**: mobilize resources and construct as per plan. *Example: excavation, foundation, frame, finishing works by the contractor.*
4. **Monitoring and control** (runs alongside execution): track progress, cost and quality against the plan, manage changes. *Example: weekly progress reviews, cube tests, earned-value checks.*
5. **Closure**: completion, inspection, handover, final bills, as-built drawings, lessons learnt. *Example: completion certificate and handover of the building.*

Q6

2 marks

> Differentiate between Agile and Waterfall project management methodologies used in construction.

	Waterfall	Agile
Approach	**Linear, sequential** phases: design → procure → build → handover	**Iterative and incremental** short cycles (sprints)

	Waterfall	Agile
Scope	Fixed and fully defined at the start	Evolves; changes welcomed
Planning	Detailed upfront plan	Rolling, adaptive planning
Client involvement	Mainly at the start and end	Continuous feedback
Change	Costly once a phase is complete	Easy to absorb
Suits	Projects with clear, stable requirements: **roads, bridges, standard buildings**	Uncertain or changing requirements: **design phase, renovation, design-build, tech-heavy fit-outs**
Risk	Problems found late	Problems found early

Q7

2 marks

> Elaborate the key roles and responsibilities of a Project Manager in a construction project.

- **Planning and scheduling**: define scope, WBS, schedule (CPM/PERT), resources.
- **Cost control**: budget, estimates, cash flow, billing and cost monitoring.
- **Quality and safety**: ensure work meets specifications and IS codes; site safety and environmental compliance.
- **Coordination and communication**: link client, consultants, contractors, suppliers; meetings and reports.
- **Procurement and contract management**: tenders, contracts, variations, claims.
- **Risk management**: identify and mitigate technical, financial and schedule risks.
- **Team leadership**: manage staff and subcontractors, resolve conflicts.
- **Monitoring and closeout**: track progress vs plan, corrective action, handover and documentation.

Q8

2 marks

> Design M15 concrete via nominal mix for fine aggregate of zone II and coarse aggregate of 20 mm size.

IS 456 Table 9 (nominal mix), **M15**: total dry aggregate = **350 kg per 50 kg of cement**; FA : CA normally **1 : 2** by mass; water = **32 litres per 50 kg cement**.

Per 50 kg bag: cement **50 kg**, FA = 350/3 = **116.7 kg**, CA = 2 × 350/3 = **233.3 kg**, water **32 l** (w/c = 0.64). Mix ratio **1 : 2.33 : 4.67 ≈ 1 : 2 : 4** by mass.

Quantities per m^3 (SG: cement 3.15, FA 2.65, CA 2.7; 2% air): volume per bag = 0.0159 + 0.0440 + 0.0864 + 0.032 = 0.1783 m^3 → 0.182 m^3 with air → 5.5 bags per m^3 → **cement ≈ 275 kg, FA ≈ 641 kg, CA ≈ 1282 kg, water ≈ 176 l**.

Q9

2 marks

> What are the three triple constraints in project management?

Scope, Time (schedule) and Cost, with **quality** at the centre. Changing one affects the others: e.g. adding scope needs more time or money; cutting time (crashing) raises cost or risks quality.

Q10

1 mark

> What is the minimum cement content for very severe exposure condition and for 10 mm size aggregate?

IS 456 Table 5 (RCC, very severe) gives **340 kg/m^3** for 20 mm aggregate. Table 6 adds **+40 kg/m^3 for 10 mm** aggregate → **380 kg/m^3** (plain concrete: 260 + 40 = 300 kg/m^3).